

MICHAEL WHITEMAN

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Education

University of Waterloo

Undergraduate in Computer Engineering

Sep. 2023 – Present

Waterloo, Ontario

Technical Skills

Languages: C, C++, C#, SystemVerilog, JavaScript, Python, ARM, VHDL, HTML/CSS

Embedded Development: Linux, Git/GitHub, Vivado, Vitis, Proteus, AutoDesk Fusion, STM32CubeIDE, Arduino, Quartus Prime, Raspberry Pi 4, RISC-V, Valgrind, JWT, ARM Cortex-M4, FreeRTOS, Visual Studio, VSCode, Windows

Hardware Skills: 3D Printing, PCB Building, Soldering, Multimeter, Oscilloscope, DMM, Function Generator

Experience

Software Developer | C++, RPI4, React, JS, JWT, MySQL, AutoDesk Fusion Jan 2026 – Apr 2026

Pinkbyte

Concord, Ontario

- Prototyped wireless **serial communication** using a Raspberry Pi 4, parsing input commands and **translating** them into **opcodes** dispatched to a HERO robot over serial interface.
- Implemented a **multithreaded alert system** in C++ to handle real-time event notifications across concurrent embedded and web application processes and integrated with 3rd party service to send those alerts to company emails.
- **Developed** secure user authentication with JWT based login and sessions for a fullstack internal web platform.
- **3D modelled** a carbon fibre RPI4 enclosure in **AutoDesk Fusion** for hardware deployment and **daily company use**.

Coding Instructor | C#, C++, JavaScript, Github, Unity, Postman API

Sept 2024 – Dec 2024

Code Ninjas

Richmond Hill, Ontario

- **Communicated complex technical topics clearly and effectively**, helping students understand advanced C++ and C# concepts while building confidence in problem-solving and logical thinking through custom coding problems.
- Utilized C++ and **Postman API** to fix bugs and developed back-end for website manager side project.
- **Demonstrated leadership and initiative** by designing new learning materials, collaborating with other instructors, and leading workshops that encouraged teamwork and hands-on creativity in 3D modeling and game creation.

Projects

Preemptive RTOS Kernel & Memory Allocator | C, ARM, STM32CubeIDE

Apr 2026 – Aug 2026

- Built a **real-time kernel** in C on **ARM Cortex-M4**, implementing task creation, **PCB/TCB management**, priority-based EDF scheduling, and **low-level context switching** via custom ARM assembly routines and PendSV.
- Wrote **low level context-switch routines** using the **PendSV exception handler** in ARM assembly, manually saving and restoring the CPU register states (R4–R11 and PSP) to **guarantee** correct task resumption.
- Created a **custom dynamic memory allocator** in C using a First-Fit algorithm, without using the standard library.
- Implemented **priority-based** and **Earliest Deadline First scheduling** to guarantee deterministic real-time task execution under strict timing constraints to meet design requirements.

Automated Water Reservoir System | C, STM32CubeIDE, Proteus, Soldering

Oct 2025 – Nov 2025

- Developed complete embedded hardware stack for a system using **STM32**, HCSR04 ultrasonic sensor, optical RPM encoder, servo-driven valve selector, L9110 motor driver, and custom PCB-mounted timer board modules.
- Utilized **Proteus** to **design a PCB** for all of the external components from the prototype into a more compact design.
- Built a real-time diagnostic output pipeline that streams hourly system-state reports, including mode, PWM %, measured RPM, water depth %, and active zone.

Embedded FPGA Audio Player | C, Altera MAX10 FPGA, Quartus Prime, NIOS II

Oct 2025 – Dec 2025

- Developed modular firmware (loopback, UART CLI, button interface, WAV player) in C using Altera Eclipse SBT.
- Employed **SignalTap II** for signal viewing and logic analyzer for timing and synchronization verification.
- **Multiplexed** dual 7-segment displays to show converted values, **improving readability on constrained hardware**.

LTE RC Car | C++, Linux, AutoDesk Fusion, Github, Raspberry Pi 4, Soldering

June 2025 – Aug 2025

- Developed a C++ client-server network architecture to **stream real-time inputs** and telemetry with low latency.
- Designed and assembled an LTE-controlled RC car using a Raspberry Pi 4 and SIM7600 module to enable **remote operation over effectively unlimited range**.
- Implemented **PWM-based motor and servo control** using custom software and hardware PWM signals.
- Applied **soldering and wiring techniques** to connect and to guarantee functionality of all electronic components.